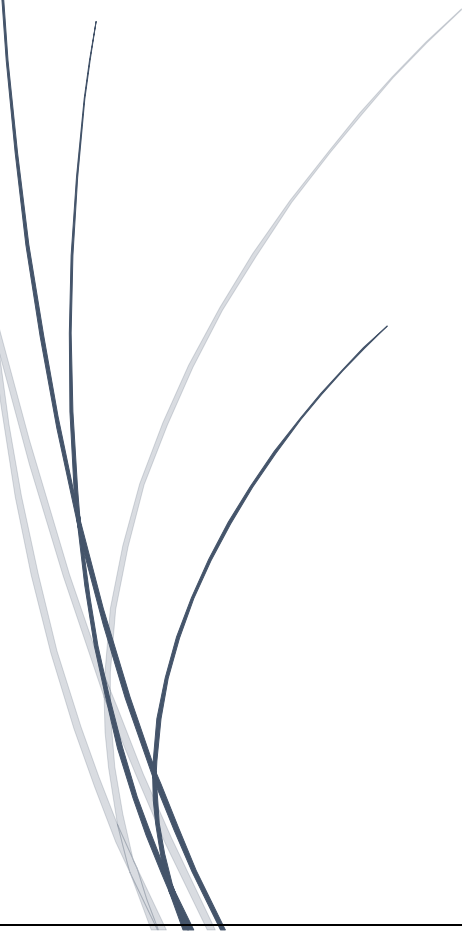




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Project Report

AI-Chatbot



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AI- Powered Data Analytics Chatbot

Protect Description:

This project is aimed at developing an AI-powered Data Analytics Chatbot which helps users. students to learn and apply Data Analytics and Data Science Techniques on various kinds of data to fetch the optimal and useful output in a matter of seconds by data relative prompts.

This bot is built using several powerful libraries such as Numpy, Pandas, Seaborn, Matplotlib. Scikit-Learn Scipy, Plotly, Streamlit. Regex, and many more. It uses LLM integration model (Gemini I.5 Flash) using the model Google Generative AI module-

This bot will assist students in generating insights analytics performing statistical analysis visualizations and major understanding core ML functions.

Tools and Technologies Used:

To perform this project Jupyter Notebook cannot be used of its limited integrations. Instead, tools like Visual Studio Code as well as PyCharm can be used to perform this project since it comes various integrations

In this project we will be moving forward the Studio Code as the IDE.

Objective:

To develop an AI-integrated, interactive chatbot that helps users perform Exploratory Data Analysis (EDA). generate visualizations. and build ML models through natural queries.

Models Used for this Project:

- Python Fundamentals.
- Numpy.
- Pandas
- Matplotlib
- Seaborn

- Scipy
- Statistics
- Sci-kit Learn
- Plotly
- Google Generative AI
- Regex
- Base64
- IO
- PIL

Key Features of This Project:

- Data upload and handling - The users can upload datasets or files. and the bot will display first few rows as a preview and generate insights based on the query provided
- EDA — Users can request statistics, visualizations, and several other Data Analysis tools or functions
- Statistical Analysis— Users can fetch descriptive statistics as well testing statistics.
- ML Assistance—The user can build or utilize basic ML models such as supervised models or unsupervised models. Also. it provides and utilizes model evaluation techniques such as accuracy classification report and many more
- Gemini Integration —If the user has some queries regarding the data, the Gemini API can be utilized as Gemini 1.5 Flash can be used to generate human like responses
- Query parsing with regex — This bot can recognize patterns structures provided by the user such as several data queries as well as statistical queries.
- Visualizations - Plotly for interactive charts, Matplotlib/Seaborn for static charts (now viewable using PIL + Base64 workaround). Supports violin plot. histogram. boxplot. scatterplot. etc.

Future Enhancements for this project:

- Memory persistence via local JSON or database (e.g., SQLite).
- User authentication and role-based access.
- Enhanced query understanding via LangChain or semantic parsers.
- Autosuggestion of visualizations from data patterns.
- Export chat and analysis as PDF or Excel reports

- Prompt Memory Saving as well as clear button
- Voice Commands for Prompts.
- Advanced ML. DL Integration.

Project File Structure:

- Data Analytics Chatbot (Folder)
 - streamlit (Folder)
 - secrets.toml
 - assets (Folder)
 - static_plots (Folder)
 - venv (Folder)
 - app.py
 - chatbot_logic.py
 - gemini_handler.py
 - utlis.py
 - requirements.txt

References:

- Python Documentation
- Streamlit Documentation
- Pandas Documentation
- Plotly Documentation
- Google AI Studio for API Key
- Gemini LLM (gemini.com)

Appendices: AI-Powered Data Analytics Chatbot

Appendix A: Project Purpose & Audience

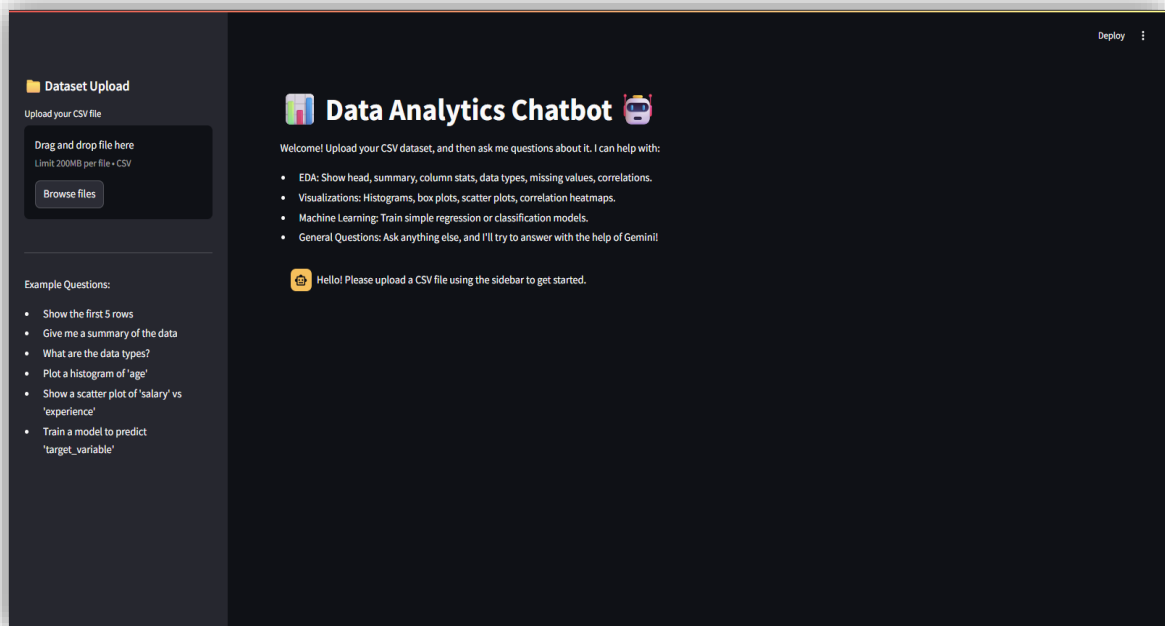
Purpose

This project was created to **democratize data analytics** by combining:

- **AI-powered natural language interfaces** (Gemini 1.5 Flash)
- **Robust Python-based data tools**
- **Interactive visual feedback** via Streamlit

It's designed for:

- **Students** learning EDA, statistics, and ML
- **Data analysts** for rapid prototyping
- **Non-technical users** who want to analyze datasets without writing code



Appendix B: LLM & Natural Language Understanding

Gemini 1.5 Flash Integration

- Used through Google AI Studio's API Key.

- Enables contextual understanding of questions like:

"Can you build a logistic regression model for this dataset?"

Regex-Based Query Parsing

- Helps detect patterns such as:
 - Column-based stats (mean, median, etc.)
 - Plot commands
 - Model selection queries (e.g., "train a decision tree")

Appendix C: Feature Workflows

1. Data Upload Workflow

- Streamlit file uploader captures .csv files.
- Automatically parses with `pandas.read_csv()`.
- Displays a preview using `data.head()`.

2. Exploratory Data Analysis

- Upon prompt like *"show me summary statistics"*, the bot performs:
 - `data.describe()`
 - `data.info()`

3. Visualizations

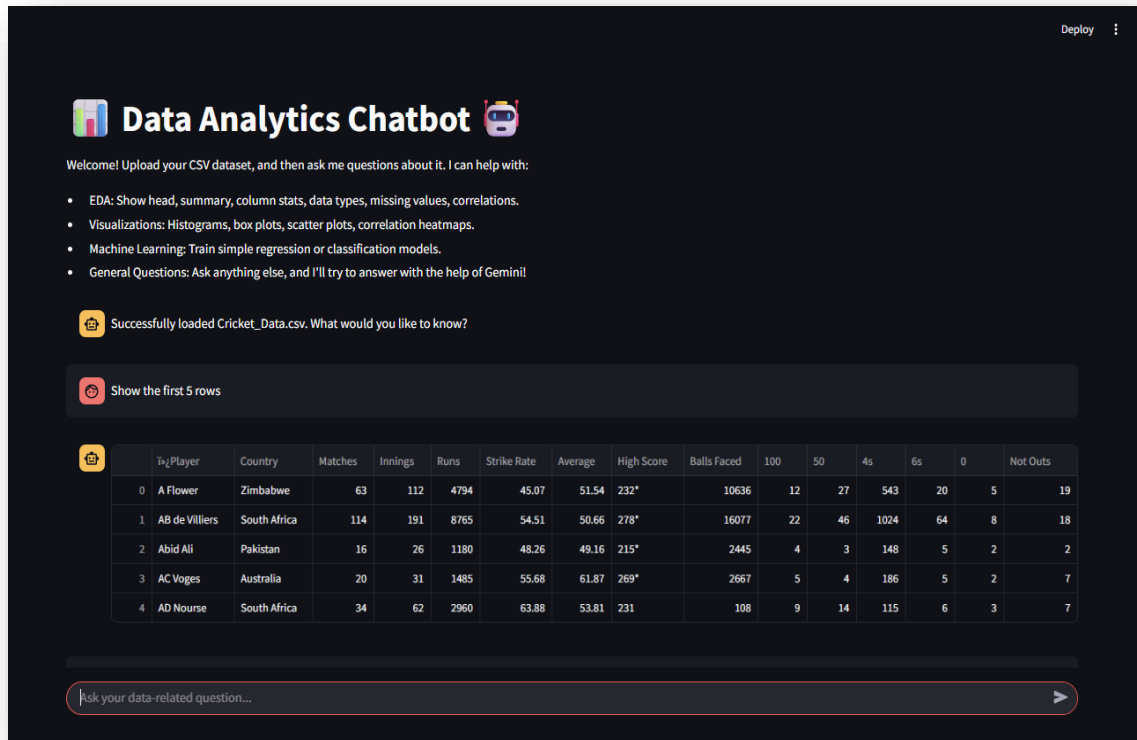
Chart Type	Libraries	Trigger Phrase Example
Histogram	Seaborn, Plotly	"histogram of age"
Boxplot	Seaborn, Matplotlib	"boxplot of salary by gender"
Violin Plot	Seaborn	"violin plot of income"
Scatter	Plotly, Matplotlib	"scatterplot of age vs income"

4. ML Model Assistance

- Uses Scikit-learn for:
 - Classification: Logistic Regression, Decision Tree

- Clustering: KMeans
- Regression: Linear Regression
- Example Query:

"Train a decision tree to predict salary"



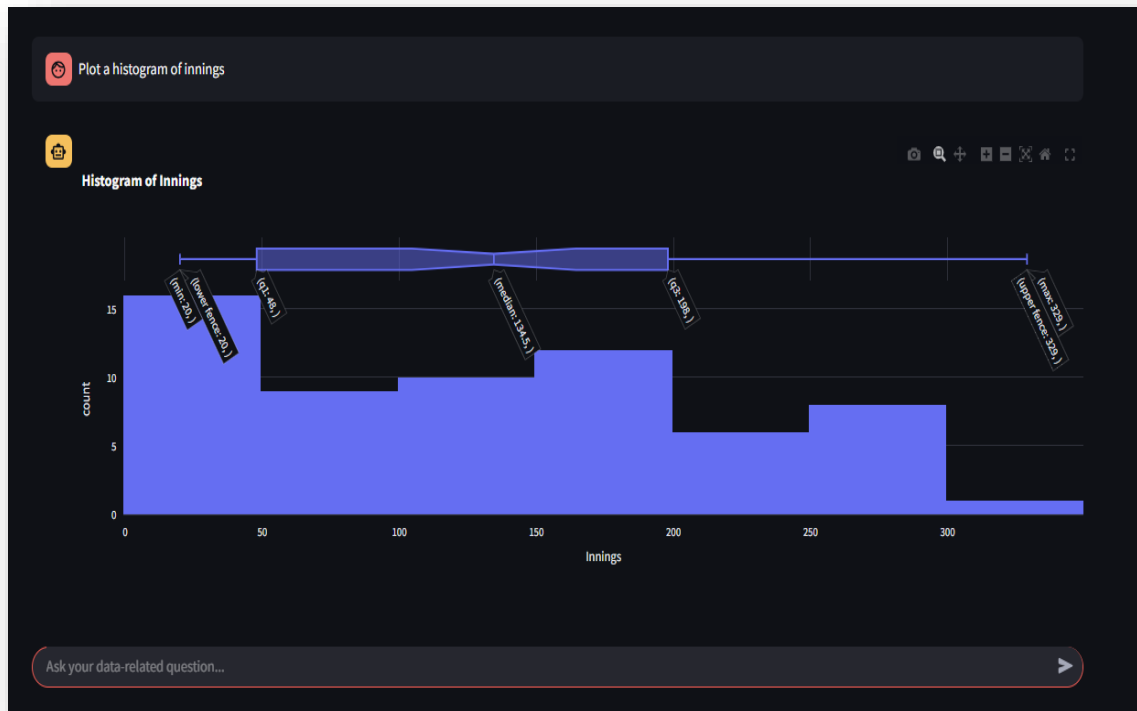
Appendix D: Key Libraries & Roles

Library	Purpose
pandas	Data manipulation and CSV handling
numpy	Numerical computing
scipy	Statistical tests and utilities
seaborn	Static statistical plots
matplotlib	Basic plotting support
plotly	Interactive plots

Library	Purpose
scikit-learn	Machine learning models
regex	Parsing structured text queries
PIL, base64	Rendering static plots in Streamlit
streamlit	Web interface for interactive analysis
google.generativeai	LLM integration with Gemini API

Appendix E: File Descriptions

File/Folder	Description
app.py	Main Streamlit app entry point
chatbot_logic.py	Logic to interpret user queries
gemini_handler.py	Handles API calls to Gemini LLM
utils.py	Helper functions (plotting, stats)
streamlit/secrets.toml	Contains Gemini API key
assets/	Images, icons, or style assets
static_plots/	Folder to store static rendered charts
venv/	Python virtual environment
requirements.txt	All dependencies to run the project



Appendix F: Limitations

- Requires internet for Gemini API calls.
- No persistent chat memory yet.
- Doesn't handle very large datasets efficiently.
- Voice input and PDF export features are not implemented yet.

Appendix G: Glossary

Term	Definition
EDA	Exploratory Data Analysis – initial investigations using summary statistics and visualizations
LLM	Large Language Model – advanced AI that understands and generates human-like text
Gemini 1.5 Flash	Google's fast and lightweight generative AI model
Regex	Regular expressions, used for pattern matching in text

Term	Definition
Classification Report	A performance metric summary for classification models
Plotly	A library for interactive, zoomable, and web-ready plots

Appendix H: Sample Prompts to Try

Prompt	Expected Output
"Show correlation matrix"	Heatmap chart
"Generate a boxplot of salary"	Boxplot visualization
"Train a KMeans clustering model"	Cluster summary with plot
"Give descriptive statistics of age"	Mean, median, std, etc.
"Predict outcome using decision tree"	Model accuracy and report